

The “Organizer” of Spemann and Mangold

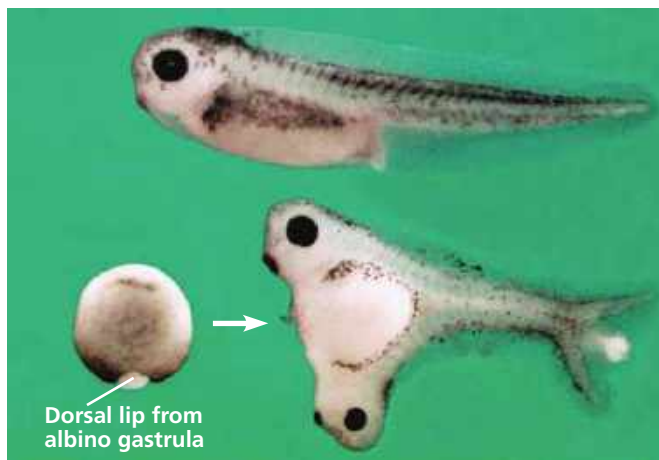
Before his studies of totipotency in the fertilized frog egg, Spemann had investigated cell fate determination during gastrulation. In these experiments, he and his student Hilde Mangold transplanted tissues between early gastrulas. In their most famous such experiment, summarized in **Figure 47.24**,

▼ Figure 47.24 Inquiry

Can the dorsal lip of the blastopore induce cells in another part of the amphibian embryo to change their developmental fate?

Experiment In 1924, Hans Spemann and Hilde Mangold, at the University of Freiburg, Germany, investigated the inductive ability of the dorsal lip of the gastrula. Using newts, they transplanted a piece of the dorsal lip from one gastrula to the ventral side of a second gastrula. Because the donor embryo was albino and thus lacked pigmentation, the researchers could visually follow how the transplanted material altered the fate of the recipient embryo.

Results The photograph in this figure documents a repeat of this classic experiment, using the frog *Xenopus laevis*. The tadpole at the top developed from a control gastrula. When an experimental gastrula received the transplant of a dorsal lip from an albino donor (lower left), the recipient embryo formed a second notochord and neural tube in the region of the transplant. Eventually most of a second embryo developed, producing a twinned tadpole (lower right).



Conclusion The transplanted dorsal lip was able to induce cells in a different region of the recipient to form structures different from their normal fate. In effect, the transplanted dorsal lip “organized” the later development of an entire extra embryo.

Data from H. Spemann and H. Mangold, Induction of embryonic primordia by implantation of organizers from a different species, *Trans. V. Hamburger* (1924). Reprinted in *International Journal of Developmental Biology* 45:13–38 (2001); and E. M. De Robertis and H. Kuroda, Dorsal-ventral patterning and neural induction in *Xenopus* embryos, *Annual Review of Cell and Developmental Biology* 20:285–308 (2004).

WHAT IF? Because the transplant caused the recipient tissue to become something it would not have otherwise, a signal must have passed from the dorsal lip. If you identified a protein candidate for the signaling molecule, how would injecting it into ventral cells of a gastrula test its function?

they made a remarkable discovery. Not only did a transplanted dorsal lip of the blastopore continue to be a blastopore lip, but it also triggered gastrulation of the surrounding tissue. They concluded that the dorsal lip of the blastopore in the early gastrula functions as an “organizer” of the embryo’s body plan, inducing changes in surrounding tissue that direct formation of the notochord, the neural tube, and other organs.

Nearly a century later, developmental biologists are still studying the basis of induction by what is now called *Spemann’s organizer*. An important clue has come from studies of a growth factor called bone morphogenetic protein 4 (BMP-4). One major function of the organizer seems to be to inactivate BMP-4 on the dorsal side of the embryo. Inactivating BMP-4 allows cells on the dorsal side to make dorsal structures, such as the notochord and neural tube. Proteins related to BMP-4 and its inhibitors are found as well in invertebrates such as the fruit fly, where they also function in regulating the dorsal-ventral axis.

Formation of the Vertebrate Limb

Inductive signals play a major role in **pattern formation**, the process governing the arrangement of organs and tissues in their characteristic places in three-dimensional space. The molecular cues that control pattern formation, called **positional information**, tell a cell where it is with respect to the animal’s body axes and help to determine how the cell and its descendants will respond to molecular signaling during embryonic development.

In Concept 18.4, we discussed pattern formation in the development of *Drosophila*. For the study of pattern formation in vertebrates, a classic model system has been limb development in the chick. The wings and legs of chicks, like all vertebrate limbs, begin as limb buds (see Figure 47.1). These bumps consist of mesodermal tissue covered by a layer of ectoderm (see Figure 47.1, inset). Each component of a chick limb, such as a specific bone or muscle, develops with a precise location and orientation relative to three axes: proximal-distal (shoulder to fingertip), anterior-posterior (thumb to little finger), and dorsal-ventral (knuckle to palm), as shown in **Figure 47.25**.

Two regions in a limb bud have profound effects on its development. One such region is the **apical ectodermal ridge (AER)**, a thickened area of ectoderm at the tip of the bud (see Figure 47.25a). Surgically removing the AER blocks outgrowth of the limb along the proximal-distal axis. Why? The AER secretes a protein signal called fibroblast growth factor (FGF) that promotes limb-bud outgrowth. If the AER is replaced with beads soaked with FGF, a nearly normal limb develops.

The second major limb-bud regulatory region is the **zone of polarizing activity (ZPA)**, a specialized block of mesodermal tissue (see Figure 47.25a). The ZPA regulates development along the anterior-posterior axis of the limb.